

Project Name: PlayOps — KK Wagh Sports Portal		Target Milestone: 50% Mid-Point Progress Stage	
Institution: KK Wagh Institute of Engg. Education & Research		Evaluation Sprint: Sprint 2 & Sprint 3 (Weeks 3–6)	
Key Delivery Areas: Core Management, QR Passes, Teams, Fixture Engine		Current Status: 50% Milestone Completed & Operational	
50% ROADMAP COMPLETED			
MILESTONE STATUS 50% COMPLETED Mid-Project Delivery Target	CUMULATIVE TASKS 69 / 137 Tasks Phases 1, 2, and 3 Finished	ENTITIES MANAGED Sports, Venues, Teams Full CRUD + Roster Engine	CORE ALGORITHM Deterministic Fixtures Knockout Brackets + Round-Robin

1. Executive Summary & Mid-Project Overview

The **50% milestone** marks the transition of PlayOps from foundational scaffolding into a fully operational management platform. Cumulative work completed through this stage covers the complete delivery of **Phase 1 (Foundation)**, **Phase 2 (Core Management)**, and **Phase 3 (Teams & Tournaments)**. Institutional administrators can now register and manage sports, catalogue campus facilities, onboard student athletes with unique cryptographic QR identity passes, form departmental teams with strict roster validations, and configure tournaments featuring automated mathematical fixture generation for both single-elimination knockout and round-robin league formats.

Mid-Project Engineering Milestone

Breakthrough Delivery: The deterministic Knockout Bracket Generator and Cyclic Round-Robin Scheduling algorithms completely eliminate manual scheduling errors, venue booking clashes, and bye assignment disputes that previously plagued campus sports operations.

2. Scope of Cumulative Work Completed up to 50%

Reaching the 50% milestone represents delivering the entire administrative core of the sports portal:

- Phase 1 (Foundation):** Validated Next.js 15, PostgreSQL (12 relational tables), Supabase Auth, and app shell.
- Phase 2 (Core Management):** Interactive CRUD consoles for sports rules and campus venues, athlete onboarding, and digital player passes with cryptographic QR codes.
- Phase 2 (On-Field Security):** Camera-based QR verification scanner validating student identity and preventing fraud.
- Phase 3 (Teams & Rosters):** Departmental team formation, roster size enforcement, and captaincy management.
- Phase 3 (Tournaments & Fixtures):** Knockout bracket algorithms with power-of-two bye logic, round-robin scheduling, and interactive visual tournament trees.

2. Detailed Cumulative Work Breakdown (Phases 1 through 3)

2.1 Sports & Facility Management Consoles (Phase 2 Deliverable)

- **Sports Management Engine:** Administrative interface to define rules, categories (Indoor/Outdoor, Team/Individual), squad size constraints (min/max players), and custom sport icons.
- **Venues & Ground Management:** Comprehensive registry of campus sports infrastructure (Main Cricket Oval, Football Turf, Synthetic Basketball Court, Badminton Arena) with capacity and real-time maintenance status.

2.2 Student Athlete Registration & Cryptographic QR Passes (Phase 2 Deliverable)

- **Student Onboarding:** Structured registration form capturing student academic details (PRN, Department, Academic Year, Emergency Contact, and Sports Preferences).
- **Digital Player ID Pass:** Automated generation of unique cryptographic QR codes embedded directly into digital Player Pass cards.
- **On-Field QR Verification Scanner:** Mobile-compatible camera scanner enabling sports marshals to scan player passes at the ground, verifying roster authenticity and preventing student impersonation.

2.3 Team Creation & Roster Management (Phase 3 Deliverable)

- **Team Lifecycle:** Creation of departmental teams with logo uploads, designated coach/mentor, and department affiliation.
- **Roster Governance:** Enforces eligibility checks, squad size limits (e.g., Cricket squad 11–16 players), and captain assignment/transfer protocols.

2.4 Tournament Engine & Automated Fixture Generation (Phase 3 Deliverable)

- **Tournament Configuration:** Configurable tournament setups supporting Knockout and Round-Robin formats, entry deadlines, and venue assignments.
- **Registration Approval Workflow:** Team tournament registration pipeline with administrative approve/reject action dialogs.
- **Knockout Bracket Algorithm:** Pure mathematical algorithm generating balanced brackets for any team count, correctly calculating byes to next power of 2 (2^n).
- **Round-Robin Fixture Engine:** Implemented circle-method scheduling generating balanced home/away game pairings across multiple rounds.
- **Visual Tournament Tree UI:** Interactive bracket visualization component rendering match pairings from Round 1 to Finals.

3. Master Deliverables Verification Matrix — Cumulative 50% Scope

Task ID	Deliverable / Component	Phase	Priority	Status
TSK-01..15	Foundation, Auth & Database Baseline	Phase 1	P0 Must-have	[VERIFIED]
TSK-16	Sports Management CRUD Interface	Phase 2	P0 Must-have	[VERIFIED]
TSK-17	Sports Rules & Squad Size Config	Phase 2	P0 Must-have	[VERIFIED]
TSK-18	Venues & Facilities Management Console	Phase 2	P0 Must-have	[VERIFIED]
TSK-19	Ground Availability Status Tracker	Phase 2	P1 High	[VERIFIED]
TSK-20	Student Player Registration Flow	Phase 2	P0 Must-have	[VERIFIED]
TSK-21	Player Profile View & Edit Portal	Phase 2	P0 Must-have	[VERIFIED]
TSK-22	Unique Cryptographic QR Pass Generator	Phase 2	P0 Must-have	[VERIFIED]
TSK-23	On-Field QR Verification Scanner	Phase 2	P1 High	[VERIFIED]
TSK-24	Team Creation & Department Linking	Phase 3	P0 Must-have	[VERIFIED]
TSK-25	Roster Management & Eligibility Checks	Phase 3	P0 Must-have	[VERIFIED]
TSK-26	Captain Assignment & Transfer Flow	Phase 3	P1 High	[VERIFIED]
TSK-27	Tournament Wizard (Knockout/League)	Phase 3	P0 Must-have	[VERIFIED]
TSK-28	Team Registration & Approval Workflow	Phase 3	P0 Must-have	[VERIFIED]
TSK-29	Knockout Bracket Algorithm with Byes	Phase 3	P0 Must-have	[VERIFIED]
TSK-30	Round-Robin Cyclic Fixture Scheduler	Phase 3	P0 Must-have	[VERIFIED]
TSK-31	Visual Bracket Tree & Schedule Views	Phase 3	P1 High	[VERIFIED]

4. Bridge to 80% Milestone — Upcoming Roadmap

- With administrative management and tournament fixtures operational, the project enters its critical live operational phase:
- Conflict-Free Match Scheduling:** Assigning venues and timeslots with automated collision detection.
 - Live Scorekeeper Console:** Multi-sport real-time scoring interfaces for Cricket, Football, Volleyball, and Kabaddi.
 - Supabase Realtime Broadcast:** Sub-second WebSocket streaming of score changes to all spectator devices.
 - Dynamic Points Table Engine:** Instant standings computation (Played, Won, Lost, Tied, NRR, Goal Difference).
 - Sports Analytics Dashboard:** Visual performance charts and department standings via Recharts.

Formal Verification & Institutional Acceptance

Prepared By: Lead Full-Stack Engineering Team PlayOps Architecture Group	Verified By: Technical Project Lead & QA Dept. of Computer Engineering	Approved By: Director of Physical Education & Sports KK Wagh Institute of Eng. Edu. & Res.
Signature: _____ Date: 05-Oct-2026	Signature: _____ Date: 05-Oct-2026	Signature: _____ Date: 05-Oct-2026